

Summer 1

1. Exponential Increasing

Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$, $\varphi(t) = 2e^{3t}$.

Check that φ is a solution of the initial Value Problem.
(YVP)

$$\begin{cases} x' = 3x \\ x(0) = 2 \end{cases}$$

Represent the corresponding solution curve and describe its long-term behaviour

$x' = 3x$ - first-order linear homogeneous differential equation / in constant coefficients

$x(0) = 2$ - initial condition

$\hookrightarrow$ initial time $\Rightarrow$

φ is a solution of this IVP iff.

$$\begin{cases} \varphi'(t) = 3\varphi(t), \forall t \in \mathbb{R} \\ \varphi(0) = 2 \end{cases}$$

$$\varphi(0) = 2 \cdot e^{3 \cdot 0} = 2 \cdot 1 = 2$$

$$\varphi'(t) = 6e^{3t}$$

$$3\varphi(t) = 3 \cdot 2e^{3t} = 6e^{3t}$$

$$\left. \begin{array}{l} \varphi(0) = 2 \cdot e^{3 \cdot 0} = 2 \cdot 1 = 2 \\ \varphi'(t) = 6e^{3t} \\ 3\varphi(t) = 3 \cdot 2e^{3t} = 6e^{3t} \end{array} \right\} \Rightarrow \begin{cases} \varphi'(t) = 3\varphi(t), \forall t \in \mathbb{R} \\ \varphi(0) = 2 \end{cases}$$

$\Rightarrow \varphi(t)$ is a solution of this IVP

solution curve = the graph of a solution

Graph of φ



$$\lim_{t \rightarrow \infty} \varphi(t) = +\infty$$

$$\lim_{t \rightarrow -\infty} \varphi(t) = 0$$

φ is strictly increasing, in fact it is exponentially increasing
 $\varphi(t) > 0, \forall t \in \mathbb{R}$

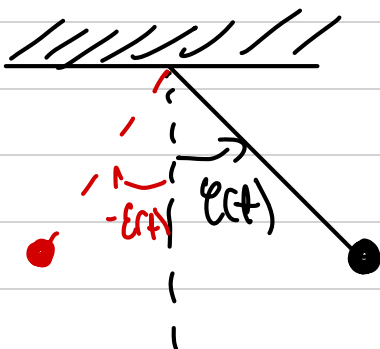
2. Oscillations w/ constant amplitude

$$\varphi(t) = \frac{1}{6} \sin t, \forall t \in \mathbb{R}.$$

Check that φ is a solution of the IVP:

$$\begin{cases} \ddot{x} + x = 0 \\ x(0) = 0 \\ \dot{x}(0) = \frac{1}{6} \end{cases} \quad \text{represent the solution curve.}$$

Describe the properties of φ .



Describe the motion of the pendulum when $\varphi(t)$ is the angle between the pendulum and the vertical.

$x'' + x = 0$ - second order linear homogeneous differential equation
with constant coefficients

→ init. position

$$x(0) = 0$$

$$x'(0) = \frac{1}{6}$$

} initial condition

↳ initial velocity

e is a solution of the IVP iff.

$$\begin{cases} e''(t) + e(t) = 0, \forall t \in \mathbb{R} \\ e(0) = 0 \\ e'(0) = \frac{1}{6} \end{cases}$$

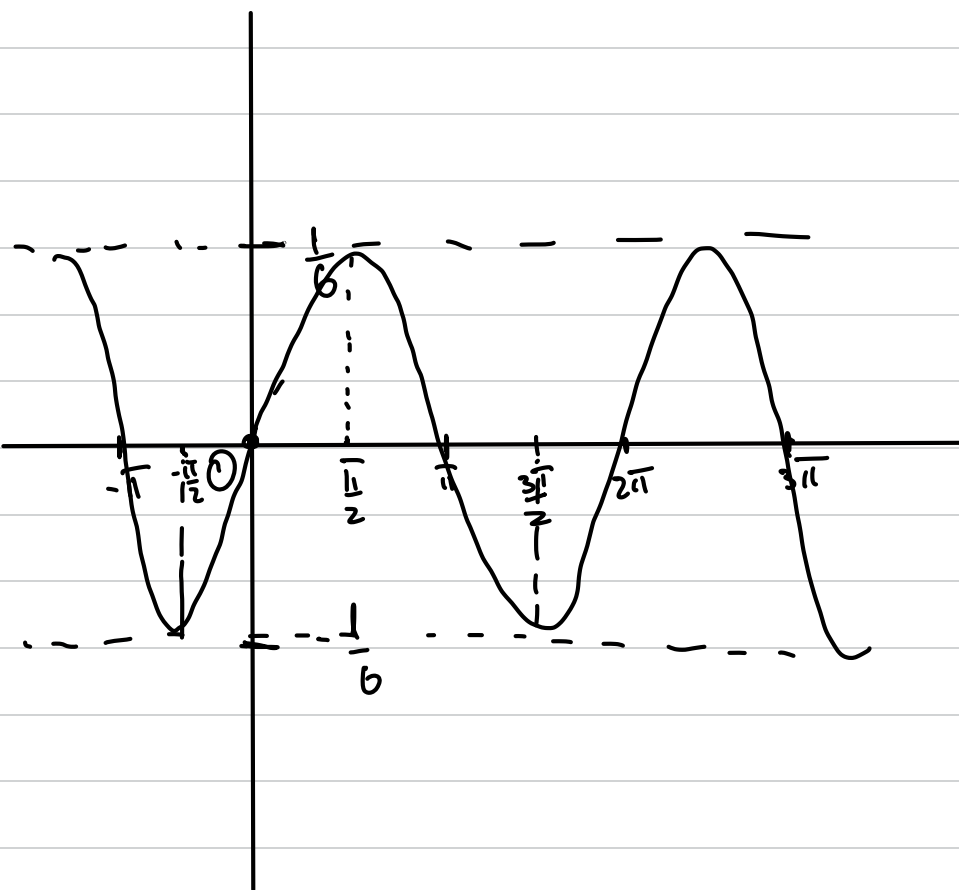
$$e(0) = \frac{1}{6} \sin 0 = 0$$

$$e'(0) = \frac{1}{6} \cos(0) = \frac{1}{6}$$

$$\begin{aligned} e'(t) + e(t) &= -\frac{1}{6} \cos t + \frac{1}{6} \sin t = \\ &= 0, \forall t \in \mathbb{R} \end{aligned}$$

} $\Rightarrow$

→ e is a solution to the IVP

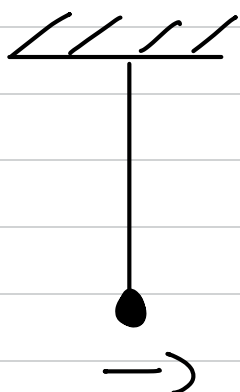


The function is bounded between $-\frac{1}{6}$, $\frac{1}{6}$, $\forall t \in \mathbb{R}$.

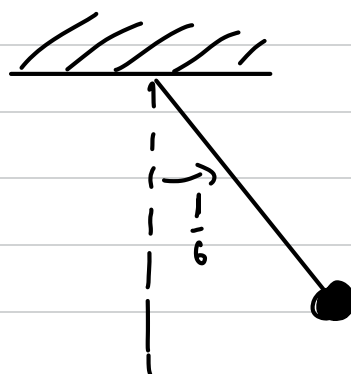
The 0's of f are $k\pi$, $\forall k \in \mathbb{Z}$

f is periodic with the main period $T = 2\pi$

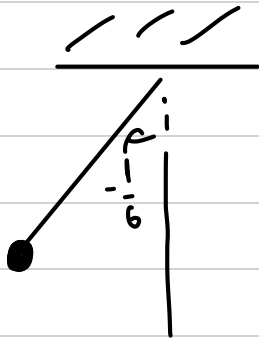
$t=0$



$t = \frac{\pi}{2} = 1, \dots$

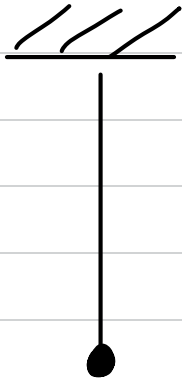


$$t = \frac{3\pi}{2} = 3, \dots \circ$$



→ oscillation with constant amplitude around a vertical

$$t = 2\pi$$



$$2, \quad e(t) = \frac{1}{6} e^{-2t} \cos t$$

Check e - solution of the IVP:

→ the term that represents the friction !!

$$\begin{cases} x'' + 4x' + 5x = 0 - \text{SHLE w/ CC} \\ x(0) = \frac{1}{6} \\ x'(0) = -\frac{1}{3} \end{cases}$$

Represent the solution conc. Describe e + the motion of the pendulum

$$\begin{aligned} e''(t) + 4e'(t) + 5e(t) &= \\ &= \frac{5}{6} e^{-2t} \cos t + \frac{4}{6} (-2e^{-2t} \cos t - e^{-2t} \sin t) + \frac{4}{6} (4e^{-2t} \cos t + 2e^{-2t} \sin t + 2e^{-2t} \sin t + e^{-2t} \cos t) \\ &= \frac{5}{6} e^{-2t} \cos t - \frac{8}{6} e^{-2t} \cos t - \frac{4}{6} e^{-2t} \sin t + \frac{16}{6} e^{-2t} \cos t + \frac{8}{6} e^{-2t} \sin t \\ &\quad + \frac{8}{6} e^{-2t} \sin t + \frac{4}{6} e^{-2t} \cos t \end{aligned}$$

$$\Rightarrow e''(t) + 4e'(t) + 5e(t) =$$

$$= -2e(t) + \frac{1}{3} e^{-2t} \sin t + e(t) =$$

$$= -2e'(t) + \frac{1}{3} e^{-2t} \sin t - e(t) - 2e(t)$$

+ ...

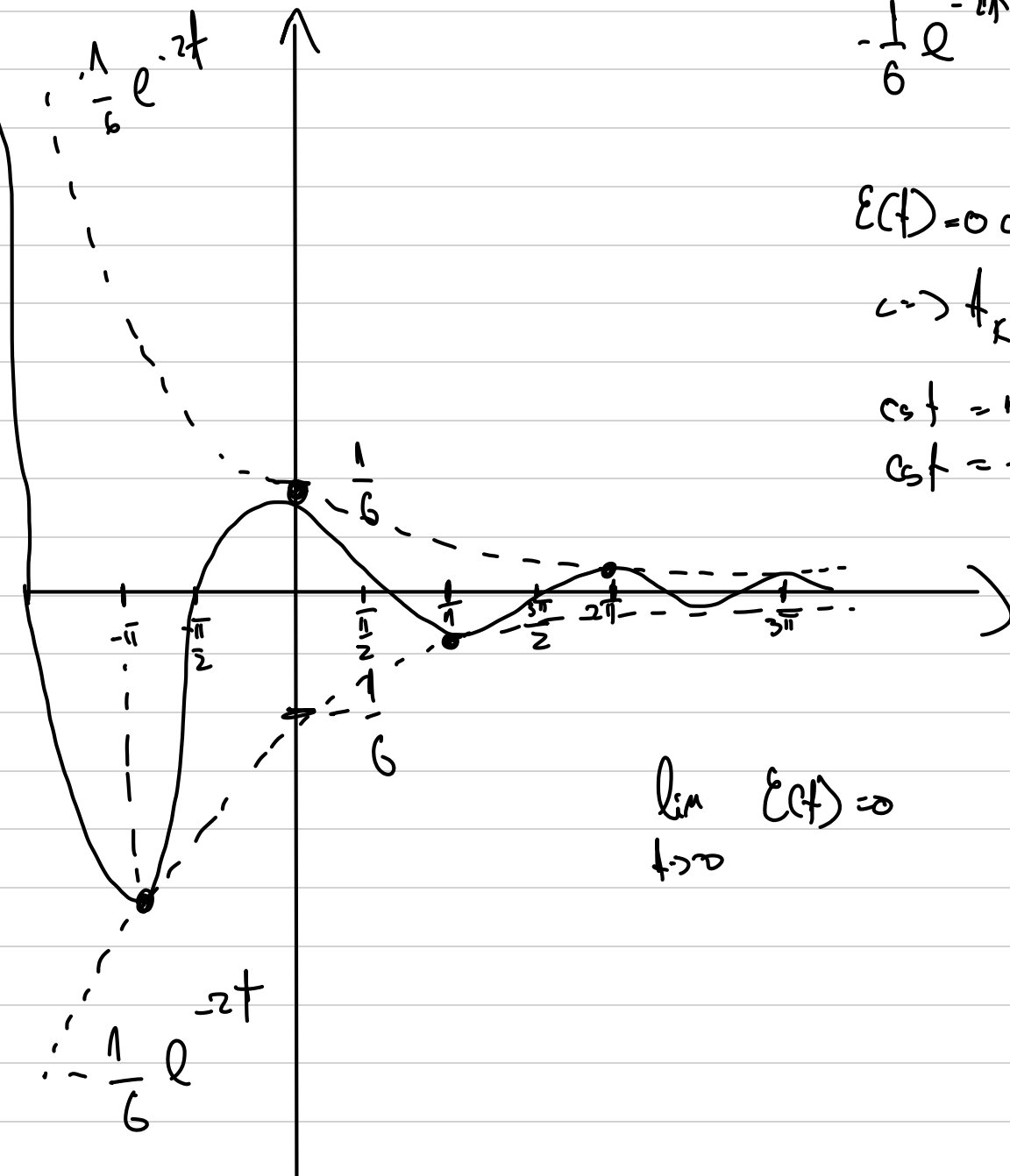
$$= \dots = 0$$

Au fait la solution n'est pas !!

$$e(0) = \frac{1}{6}$$

$$e'(0) = \frac{1}{6} (-2e^0 \cos 0 - e^0 \sin 0) = -\frac{1}{3}$$

$\Rightarrow$ A solution



$$-\frac{1}{6} e^{-2t} \leq E(t) \leq \frac{1}{6} e^{2t}$$

$$E(t) = 0 \Leftrightarrow \cos(t) = 0 \Rightarrow$$

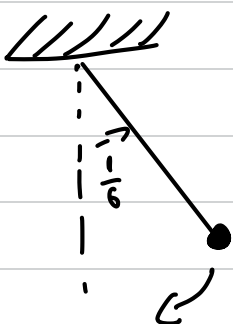
$$\Leftrightarrow t_k = k\pi + \frac{\pi}{2}$$

$$\cos t = 1 \Rightarrow t = 2k\pi$$

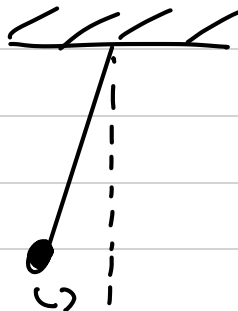
$$\cos t = -1 \Rightarrow t = (2k+1)\pi$$

$$\lim_{t \rightarrow \infty} E(t) = 0$$

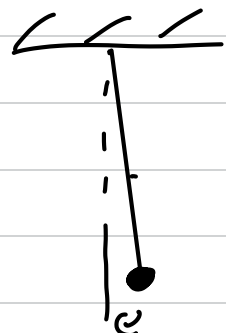
$$t=0$$



$$t = \pi = 3.14 \dots$$



$$t = 2\pi = 6.28 \dots$$



smaller and smaller angle $\Rightarrow$ OSCILLATION w/

EXPONENTIALLY DECREASING AMPLITUDE

ivp:

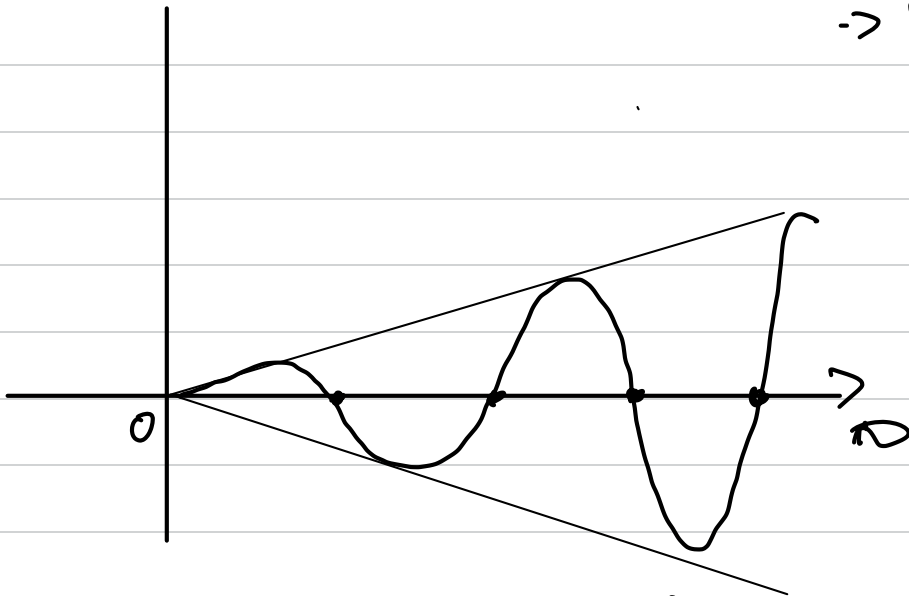
$$\begin{cases} x'' + x = 2 \sin t \\ x(0) = 0 \\ x'(0) = 0 \end{cases}$$

$$f(t) = t \sin t$$

HW: ϵ -solution

$$x'' + x = \underbrace{2 \sin t}_{\text{force}} \quad 2^{\text{nd}} \text{ order L. m-H DE}$$

-> RESONANCE



$$-t \leq f(t) \leq t, \quad \forall t \in [0, \infty)$$

$$16) \quad x_1, x_2, x_3 : \mathbb{R} \rightarrow \mathbb{R}, \quad x_1(t) = \cos(t) \\ x_2(t) = \sin(t) \\ x_3(t) = e^t$$

i) Prove that they are lin. indep. in the linear space over $\mathbb{R}$ of continuous functions defined on $\mathbb{R}$

ii) Find a solution of the form

$$x(t) = a \cos t + b \sin t + c e^t, (a, b, c \in \mathbb{R})$$

for the DE : $x' + 2x = -3 \cos t + 4e^t$

i) By definition x_1, x_2, x_3 - lin. indep. when

$$h_1 x_1 + h_2 x_2 + h_3 x_3 = 0 \Rightarrow h_1 = h_2 = h_3 = 0$$

$\hookrightarrow$ linear combination

Let $h_1, h_2, h_3 \in \mathbb{R}$ be s.t. $h_1 x_1 + h_2 x_2 + h_3 x_3 = 0 \Rightarrow$

$$h_1 \cos t + h_2 \sin t + h_3 e^t = 0, \forall t \in \mathbb{R} \Rightarrow$$

$$\Rightarrow \text{For } t=0$$

$$h_1 + h_3 = 0$$

$$\text{For } t = \frac{\pi}{2}:$$

$$h_2 + h_3 e^{\frac{\pi}{2}} = 0$$

$$\text{For } t = \pi:$$

$$-h_1 + h_3 e^{\pi} = 0$$

$$\begin{cases} h_1 + h_3 = 0 \\ \dots \\ \dots \end{cases} \quad \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & e^{\frac{\pi}{2}} \\ -1 & 0 & e^{\pi} \end{vmatrix} = e^{\pi} + 1 \neq 0 \Rightarrow$$

$$\Rightarrow h_1 = h_2 = h_3 = 0$$

the ONLY solution

$$\text{ii) let } \vec{e}(t) \text{ - solution } \Rightarrow$$

$$\Rightarrow \vec{e}'(t) + 2\vec{e}(t) = -3\vec{e}_3 t + 7\vec{e}^t$$

$$\vec{e}(t) = a \cos t + b \sin t + c e^t$$

$$\vec{e}(t) = -a \sin t + b \cos t + c e^t$$

$$\Rightarrow -a \sin t + b \cos t + c e^t + 2a \cos t + 2b \sin t + 2c e^t = -3 \cos t + 4 e^t$$

$$\Rightarrow \cos t (b+2a) + \sin t (2b-a) + 3c e^t = -3 \cos t + 4 e^t$$

$$\Rightarrow \begin{cases} b+2a = -3 \rightarrow b+4b = -3 \Rightarrow b = -\frac{3}{5} \\ 2b-a = 0 \Rightarrow a = 2b \Rightarrow a = -\frac{6}{5} \\ 3c = 4 \Rightarrow c = \frac{4}{3} \end{cases}$$

$$\Rightarrow f(t) = -\frac{6}{5} \cos t - \frac{3}{5} \sin t + \frac{4}{3} e^t$$